

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

MECHANICAL PROPERTIES OF FLUIDS / Solid

LECTURE-08 PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

1

Revision ✓

2

Gabbar Numerical Series of Mechanical Properties of Solids

2

3

Gabbar Numerical Series of Mechanical Properties of Fluids

1

4

..

RECAP FROM 1ST LECTURE

Fluid

liquid

gas

Density ρ

$$\rho = \frac{m}{V}$$

Pressure

$$P = \frac{F}{A}$$

Atmospheric pressure P_0

Variation of pressure with depth



$$P_B = P_A + \rho gh$$

2026

Imp

Pascal Law

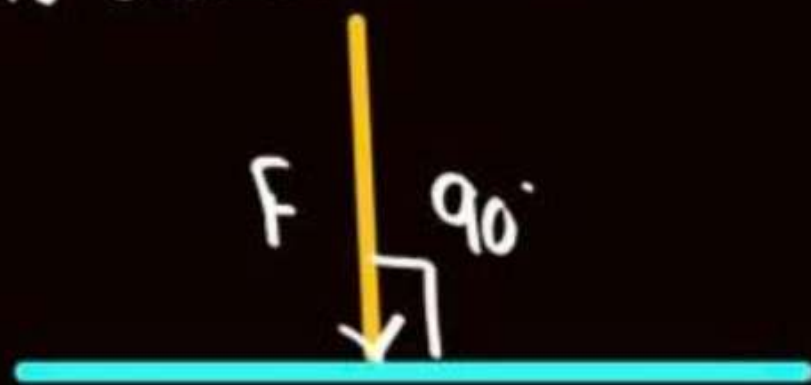
Recap-2

① Application pascal law

- Hydraulic Brake ✓
- Hydraulic lift

② Thrust

- Any perpendicular force is called thrust

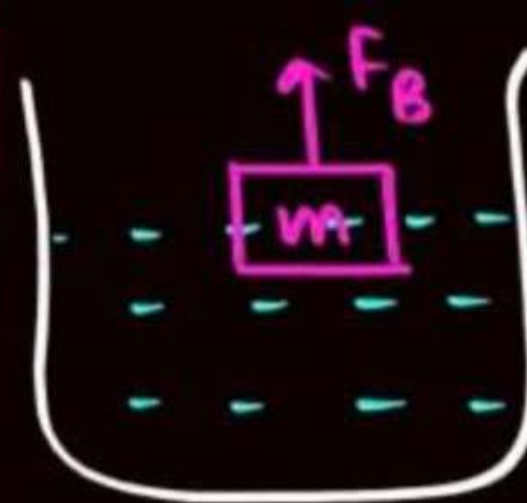


Buoyant force



Archimedes principle

$$F_B = \rho V g$$



Flow

Streamline flow

$$Re < 2000$$

Equation of Continuity

$$a_1 v_1 = a_2 v_2$$

turbulent flow

$$Re \gg \gg 2000$$

Recep - ③

Bernoullies Theorm

- law of conservation of Energy

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

Application of B-Theorm

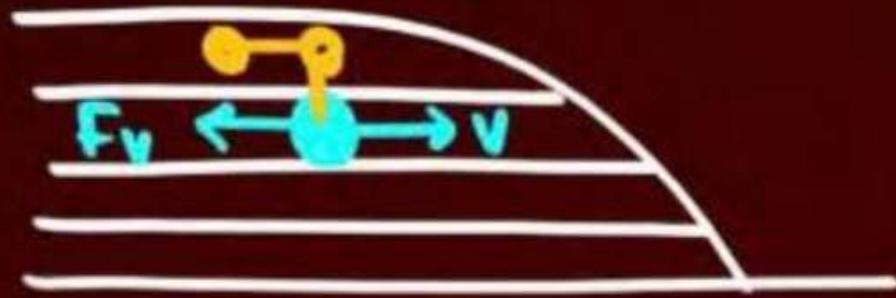
① Spray / Atomizer

② Magnus effect

③ lift of aeroplane

RECAP FROM 4TH LECTURE

• Viscosity



$$F_v = -\eta A \frac{dv}{dx} \quad \checkmark$$

$\eta \Rightarrow$ coefficient of viscosity

$\eta = \text{Pascal sec (SI)} \quad \checkmark$

$\eta = \text{poise (c.g.s)} \quad \checkmark$

Dimensional formula of η

$$\checkmark \eta = m l^{-1} t^{-1} \quad \checkmark$$

Stoke law

$$F_v = 6\pi\eta r v$$

$a=1$ (dimension)
 $b=1$
 $c=1$

RECAP FROM 5TH LECTURE



- Terminal velocity



$$v_T = \frac{2}{9} \frac{r^2 g (\rho - \rho_f)}{\eta}$$

RECAP FROM 6TH LECTURE

Application of Surface tension

(a) Shape of drop



(b) Shape of water surface



Note

Angle of Contact

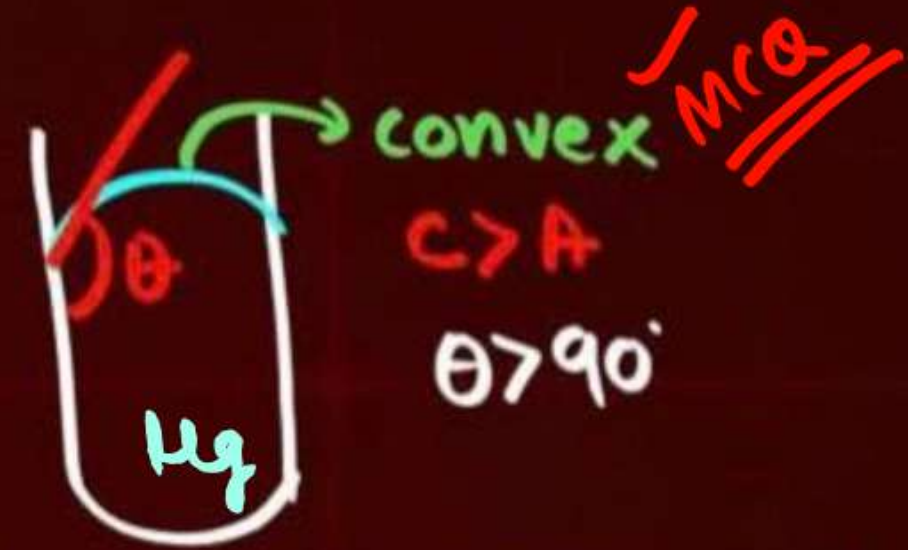
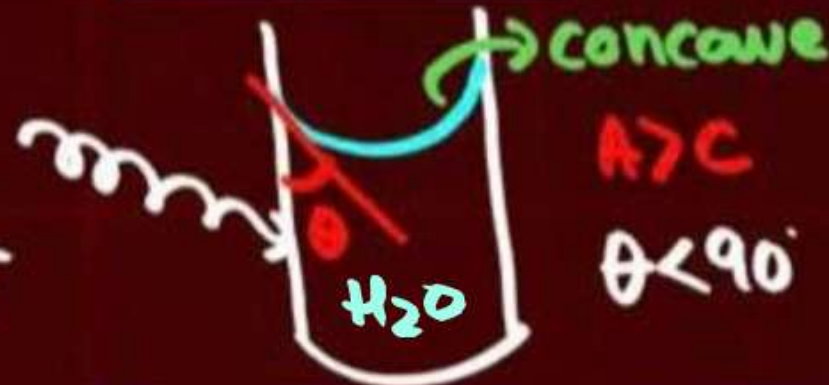
Adhesion

water + wall

cohesion

water + water

Meniscus



Excess pressure

$$P = P_0 + \rho g h$$

Effect of $\uparrow$ g on pressure

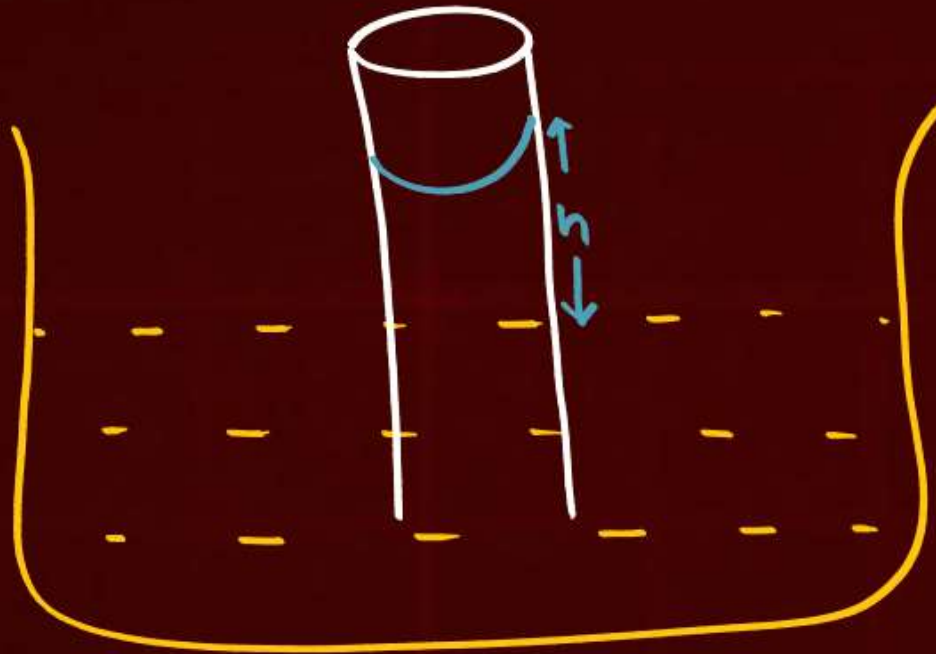
drop

$$\text{drop } P_p = \frac{2T}{R}$$

$$\text{bubble } \textcircled{O} = \frac{4T}{R}$$

RECAP FROM 7TH LECTURE

Capillary Rise



$$h = \frac{2T \cos \theta}{\rho g h}$$

Ascent formula.

Critical velocity

- ✓ The velocity when stream line flow change into turbulent flow

Chapter the end
❤️

Gabbbar

Numerical Series

PYQs ✓

Mechanical properties of Fluids ❤️

Question-01 [Summer 2026 SET-A]



3M ✓

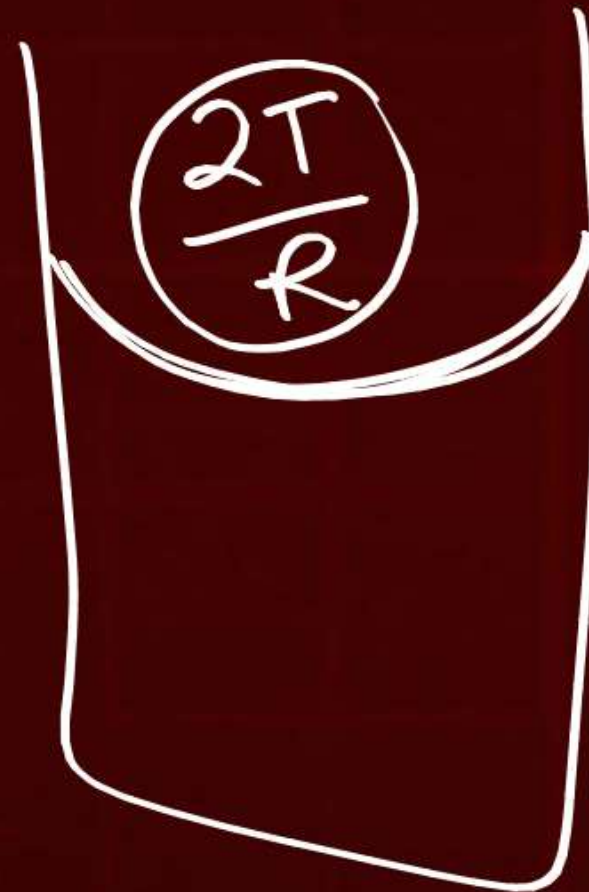
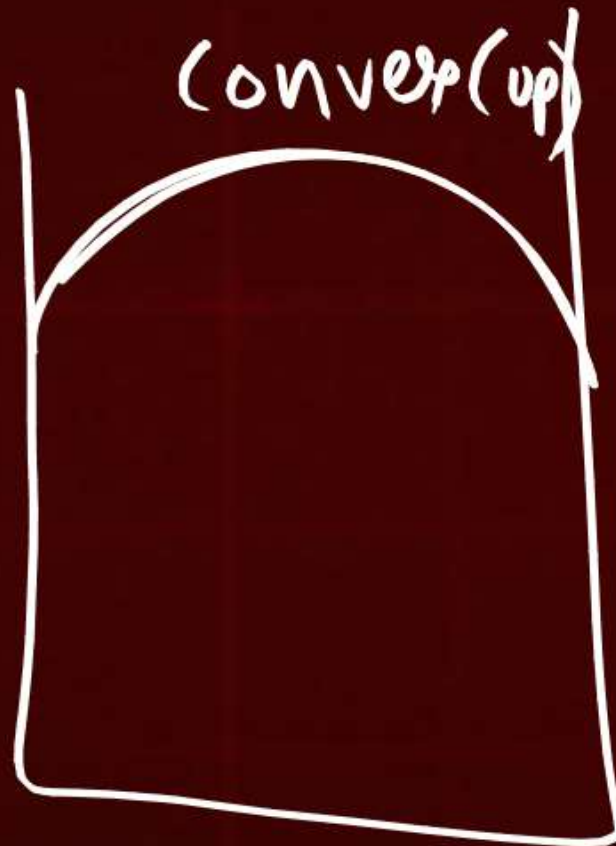
State and pascal's law and discuss hydraulic brakes on its basis.

hQ

Question-02 [Summer 2026 SET-A]

State Bernoulli's theorem and discuss its physical significance. Give two applications of Bernoulli's theorem.

5 Mark



$$EP = \frac{2T}{R}$$

Question-03 [Winter 2025 SET-A]

The meniscus of mercury in a glass tube is:

-  Concave upward
-  Convex upward
-  Plane
-  None of above

Question-04 [Winter 2025 SET-A]

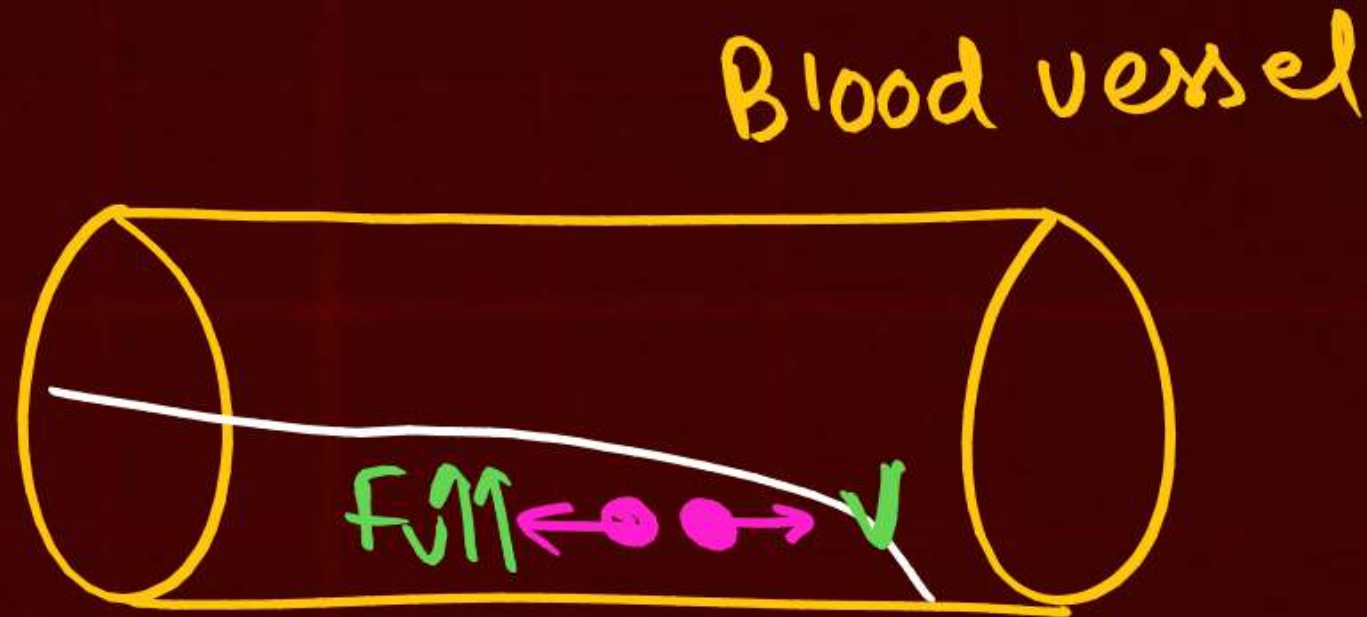
Define surface energy. Write relation between surface tension and surface energy.

$$SE = ST \times Area$$

Question-05 [Winter 2025 SET-B]

The normal circulation of blood through our arteries and veins depend on:

- A** Surface tension *✓*
- B** Capillarity *✓*
- C** Viscosity of blood *✓*
- D** None of them *✓*



Question-06 [Winter 2025 SET-B]

Legible writing on a newspaper with a fountain pen is difficult because of:



Inertia ✗



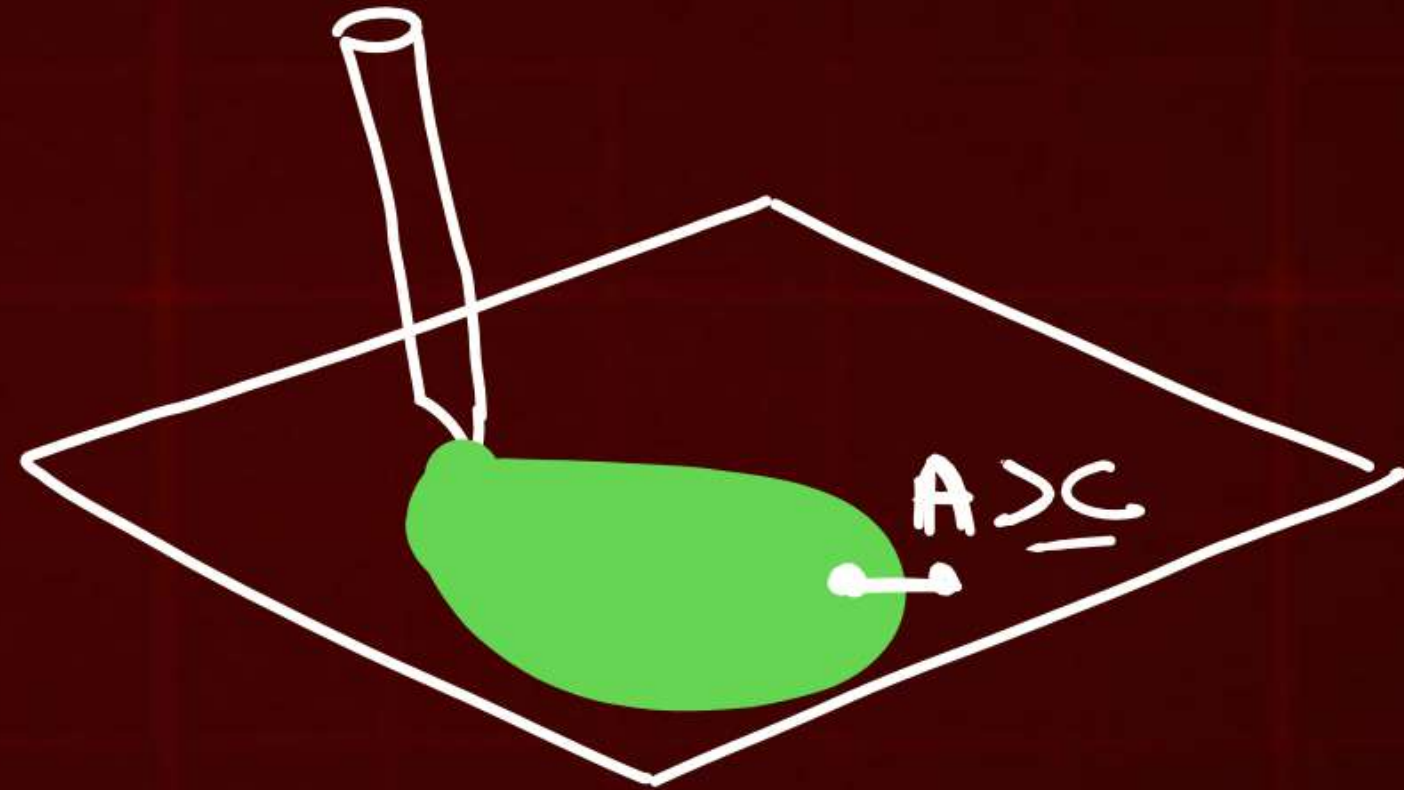
Elasticity ✗



Capillary action ✓



Surface Tension ✗



Question-07 [Winter 2025 SET-B]

Obtain the dimensions and units of coefficient of viscosity. $\eta = \text{ML}^{-1}\text{T}^{-1}$

Question-08 [Summer 2025 SET-A]

State and prove Bernoulli's theorem. Give its applications. ✓

5m

Or

Define Capillarity. Derive ascent formula for the rise of a liquid in a capillary tube and discuss the result obtained. ✓

$$AF \quad h = \frac{2T \cos \theta}{r \rho g}$$

Question-09 [Summer/Winter 2025 SET-A]

Define streamline and turbulent flow.

2M

Question-10 [Summer/Winter 2025 SET-A]

3m

State Bernoulli's theorem in different forms. Explain its one application. ✓

MECHANICAL PROPERTIES OF SOLIDS

Question-01 [JAMMU 2026 SET-A]

Which of the following is defined by Hook's law?

A

Modulus of elasticity

B

Stress

C

Strain

D

Elastic limit

Stress $\propto$ Strain

Stress = e Strain

Question-02 [JAMMU 2026 SET-B]

Which of these materials is unaffected by an applied elastic force?

A

Copper

B

Quartz

C

Steel

D

Rubber

(more Elastic)

Quartz $\rightarrow F$

$\rightarrow F$

Cu

Cu $\rightarrow F_{ext}$

Steel

Steel

Rubber

Rubber

Question-03 [JAMMU 2026 SET-C]

If the temperature increases, what happens to the modulus of elasticity?



Increases



Decreases ✓



Becomes zero



Remains constant

$\infty \rightarrow \text{femt}$

$0 \leftarrow \text{RF}$

Question-04 [Winter 2025 SET-A]

State and explain Hooke's law. ✓

Question-05 [Winter/Summer 2024 SET-A]

The dimensional formula for the modulus of elasticity is same as that of:



Stress



Strain



Velocity



Surface tension

$$e = \frac{\text{Stress}}{\text{Strain}}$$

$$e = \frac{\text{Stress}}{1}$$

$$\text{Stress} = \frac{F}{A} = \frac{MLT^{-2}}{L^2}$$

$$F = ML^{-1}T^{-2}$$

$$\underline{\underline{\text{Strain}}} = \frac{\Delta L}{L} = \frac{L}{L} = 1$$

Question-06 [Winter/Summer 2024 SET-B]

The work done in stretching an elastic wire per unit volume in a stretched wire is:



Stress × Strain



$\frac{1}{2}$ Stress × Strain ✓



$\frac{\text{Stress}}{\text{Strain}}$



None of these

$$\frac{W}{V} = \frac{1}{2} \text{ Stress} \times \text{Strain}$$

$$\frac{U}{V} = \frac{1}{2} \text{ Stress} \times \text{Strain}$$

SUMMARY



GNS — Fluid

GNS — Solid

HOMework



- copy → check
- qns → practice

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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Thank
You

